Homework 6

Q1A With the sample size provided, the minimum is 102. The next smallest is 115.

That means if 102 is removed [Jackknife is LOOCV], minimum is 115. For the rest 9 that are removed, 115 is the smallest

$\hat{\theta}_{(i)} = (115, 102, 102, 102, 102, 102, 102, 102, 102)$

The equation for the Jackknife estimate is the following:

$\widehat{\theta_{jack}} = n\bar{\theta}_{(\cdot)} - (n-1)\hat{\theta}$ Where:

$\hat{\theta} = 102$

$\bar{\theta}_{\cdot} = [(115 + (9 \cdot 102))/10] = 103.3$

Plugging into the equation

$\widehat{\theta_{jack}} = 10(103.3) - 9(102)$

$= 1033 - 918$

$= \cancel{105}\ 115$

The following are the ranges for each vocal group

Voice type	Ranges (Hz)
Bass	82-110
Baritone	110-130
Tenor	130-150
Alto	160-200
Soprano	200-250

- The Jackknife estimate of 115 Hz best fits along the Baritone voice type

$X_{(1)}$ = Smallest Observation (102)
$X_{(2)}$ = 2nd Smallest observation (115)

Plugged in Jacknife equation is the following

$$\bar{\theta}_{(\cdot)} = \frac{X_{(2)} + (n-1)X_1}{n}$$

$$\hat{\theta}_{jack} = n\bar{\theta}_{\cdot} - (n-1)X_1$$

With the numbers plugged in, $X_2 = 115$, Confirming our results

Homework6

Kunaal Raghav

Install Packages

```
#install.packages("bootstrap")
library(bootstrap)
```

```
Warning: package 'bootstrap' was built under R version 4.5.2
```

```
library(MASS)
```

Question 1: Bases and Sopranos

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

Part B

```
#Data creation
x <- c(102,115,127,127,162,180,184,205,239,240)


theta_hat <- min(x)

#LOOCV Calculation
theta_i <- sapply(1:length(x), function(i) min(x[-i]))

#Equation
theta_jack <- length(x) * mean(theta_i) - (length(x) - 1) * theta_hat

#Results
theta_jack
```

```
[1] 115
```

Note for the equation (which is theta-hat) the equation is as follows:

$$\hat{\theta}jack = n\hat{\theta} - (n-1)\bar{\theta}_{(-)}$$

Question 2: Example 5.4.9

Part A

```r
mu_hat <- mean(Boston$medv)
mu_hat
```

```
[1] 22.53281
```

Part B

```r
n <- length(Boston$medv)
se_formula <- sd(Boston$medv) / sqrt(n)
se_formula
```

```
[1] 0.4088611
```

Part C

```r
set.seed(1)

B <- 2000
boot_means <- numeric(B)

for (b in 1:B) {
  sample_b <- sample(Boston$medv, size = n, replace = TRUE)
  boot_means[b] <- mean(sample_b)
}

se_boot <- sd(boot_means)
se_boot
```

```
[1] 0.4026346
```

Both calculations are similar in response. The small difference represent sampling variability

Part D

```
ci_boot <- c(mu_hat - 2*se_boot, mu_hat + 2*se_boot)
ci_boot
```

```
[1] 21.72754 23.33808
```

```
t.test(Boston$medv)
```

```
    One Sample t-test

data:  Boston$medv
t = 55.111, df = 505, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 21.72953 23.33608
sample estimates:
mean of x
 22.53281
```

Part E

```
M_hat <- median(Boston$medv)
M_hat
```

```
[1] 21.2
```

Part F

```
set.seed(1)

boot_medians <- numeric(B)

for (b in 1:B) {
  sample_b <- sample(Boston$medv, size = n, replace = TRUE)
  boot_medians[b] <- median(sample_b)
}

se_median_boot <- sd(boot_medians)
se_median_boot
```

```
[1] 0.3788984
```